

Seed 3, Layer 2 (Round 2): The H -recursion, polynomiality of h_m , reduction to monotonicity, and a complete proof for $d = 2$

Seed 3 / Layer 2

2026-07-04

Setup

Fix $d \geq 1$ and let $\mathcal{C}_d = \{c = (c_0, c_1, c_2) \in \mathbb{Z}_{\geq 0}^3 : c_0 + c_1 + c_2 = d\}$ be the rank-3 profiles. For $c', c \in \mathcal{C}_d$ put $e = c' - c$ and

$$\text{EMD}(c', c) = 2e_0 + e_1 + 3 \max(0, -e_0, -e_0 - e_1) \in \mathbb{Z}_{\geq 0}.$$

Define $P_{c,0} = 1$ and $P_{c,m} = \sum_{c' \in \mathcal{C}_d} q^{m \text{EMD}(c', c)} P_{c', m-1}$, and $F_{c,m} = P_{c,m} / (q^3; q^3)_m$, $g_{c,m} = F_{c,m} - F_{c, m-1}$, $h_{c,m} = (q; q)_m g_{c,m}$ (case $\gcd(d, 3) = 1$, so $\ell = 1$). The Layer-1 bottleneck is: $h_{c,m} \geq 0$ coefficientwise.

1 The H -recursion (exact computation)

Theorem 1 (H -recursion). *Let $H_{c,m} := (q; q)_m F_{c,m}$. Then $H_{c,0} = 1$ and for $m \geq 1$*

$$(1 + q^m + q^{2m}) H_{c,m} = \sum_{c' \in \mathcal{C}_d} q^{m \text{EMD}(c', c)} H_{c', m-1}, \quad (1)$$

and moreover

$$h_{c,m} = H_{c,m} - (1 - q^m) H_{c, m-1} = (H_{c,m} - H_{c, m-1}) + q^m H_{c, m-1}. \quad (2)$$

Proof. From $F_{c,m} = P_{c,m} / (q^3; q^3)_m$ and the transfer recursion, $(q^3; q^3)_m F_{c,m} = \sum_{c'} q^{m \text{EMD}(c', c)} (q^3; q^3)_{m-1} F_{c', m-1}$. Divide by $(q; q)_{m-1}$ and use $(q^3; q^3)_m / (q^3; q^3)_{m-1} = 1 - q^{3m}$ and $(q; q)_m / (q; q)_{m-1} = 1 - q^m$:

$$\frac{1 - q^{3m}}{1 - q^m} H_{c,m} = (1 + q^m + q^{2m}) H_{c,m} = \sum_{c'} q^{m \text{EMD}(c', c)} H_{c', m-1}.$$

For (2): $h_{c,m} = (q; q)_m (F_{c,m} - F_{c, m-1}) = H_{c,m} - (1 - q^m) (q; q)_{m-1} F_{c, m-1} = H_{c,m} - (1 - q^m) H_{c, m-1}$. $\square$

Remark 2. Equation (1) allows exact computation in $\mathbb{Z}[q]$ (division with remainder by $1 + q^m + q^{2m}$, asserting zero remainder), eliminating all power-series truncation issues that plagued Round 1.

2 Polynomiality of $H_{c,m}$ and $h_{c,m}$ when $\gcd(d, 3) = 1$

Let $\rho(a_0, a_1, a_2) = (a_1, a_2, a_0)$ be profile rotation.

Lemma 3 (Rotation shift). *For all $c', c \in \mathcal{C}_d$: $\text{EMD}(\rho c', c) \equiv \text{EMD}(c', c) + d \pmod{3}$.*

Proof. Modulo 3, $\text{EMD}(c', c) \equiv 2e_0 + e_1 = (2c'_0 + c'_1) - (2c_0 + c_1)$ (the max term is a multiple of 3). Replacing c' by $\rho c'$ changes $2c'_0 + c'_1$ to $2c'_1 + c'_2 \equiv 2c'_1 + c'_2 + 3c'_0 - (2c'_0 + c'_1) + (2c'_0 + c'_1) = (2c'_0 + c'_1) + (c'_0 + c'_1 + c'_2) \pmod{3}$, i.e. adds d modulo 3. $\square$

Lemma 4 (Rotation equivariance). $\text{EMD}(\rho a, \rho b) = \text{EMD}(a, b)$, hence $P_{\rho c, m} = P_{c, m}$, $F_{\rho c, m} = F_{c, m}$, $H_{\rho c, m} = H_{c, m}$ (verified symbolically and numerically).

Theorem 5 (Polynomiality). *If $\gcd(d, 3) = 1$, then $H_{c, m} \in \mathbb{Z}[q]$ and $h_{c, m} \in \mathbb{Z}[q]$ for all $c \in \mathcal{C}_d$, $m \geq 0$.*

Proof. Induction on m ; $H_{c, 0} = 1$. Since $\gcd(d, 3) = 1$ and ρ has order 3, every ρ -orbit $O \subset \mathcal{C}_d$ is free: a fixed point requires $c_0 \equiv c_1 \equiv c_2$, forcing $3 \mid d$. Group the right side of (1) into orbits $O = \{a, \rho a, \rho^2 a\}$. By Lemma 4 all three members share $H_{O, m-1} := H_{a, m-1}$, and by Lemma 3 their exponents $m \text{EMD}(\cdot, c)$ hit the three residues $m \text{EMD}(a, c)$, $m(\text{EMD}(a, c) + d)$, $m(\text{EMD}(a, c) + 2d)$ modulo $3m$; since $\gcd(md, 3m)$ -classes $\{0, md, 2md\}$ cover $\{0, m, 2m\} \pmod{3m}$ when $\gcd(d, 3) = 1$, each orbit contributes $H_{O, m-1} q^{m\alpha_O} (1 + q^{m\beta} + q^{2m\beta})$ -type sums congruent to $H_{O, m-1} (x^{a_0} + x^{a_1} + x^{a_2})$ with $x = q^m$ and $\{a_i\}$ covering all residues mod 3. Each such trinomial is divisible by $1 + x + x^2$ in $\mathbb{Z}[x]$ (as $x^a \equiv x^{a \bmod 3} \pmod{1 + x + x^2}$ and $1 + x + x^2 \mid x^0 + x^1 + x^2$). Hence the whole right side of (1) is divisible by $1 + q^m + q^{2m}$ in $\mathbb{Z}[q]$, so $H_{c, m} \in \mathbb{Z}[q]$; then $h_{c, m} \in \mathbb{Z}[q]$ by (2). $\square$

Remark 6. *For $3 \mid d$ the orbit $(d/3, d/3, d/3)$ is fixed and the division fails already at $m = 1$ (verified for $d = 3$), consistent with the known failure of $h_m \geq 0$ there. This answers attack line (c): divisibility of $\tilde{h}_m$ by $\prod_i (1 + q^i + q^{2i})$ always holds, because $\tilde{h}_m$ is $h_{c, m}$ expressed through H .*

3 Reduction of $h_m \geq 0$ to monotonicity

Proposition 7 (Reduction). *If $H_{c, m} \geq H_{c, m-1}$ coefficientwise for all c, m (Monotonicity), then $H_{c, m} \geq 0$ and $h_{c, m} \geq 0$ coefficientwise for all c, m .*

Proof. $H_{c, 0} = 1 \geq 0$ and monotonicity give $H_{c, m} \geq 0$ by induction. Then (2) writes $h_{c, m}$ as a sum of two coefficientwise-nonnegative terms. $\square$

Conjecture 8 (Monotonicity). *For $\gcd(d, 3) = 1$: $H_{c, m} \geq H_{c, m-1}$ coefficientwise for all $c \in \mathcal{C}_d$, $m \geq 1$.*

Evidence. Verified *exactly* in $\mathbb{Z}[q]$ for $d \in \{2, 4, 5, 7, 8, 10, 11\}$, all profiles, $m \leq 5$ –6; and $h_{c, m} \geq 0$ verified exactly for $d = 13$ (105 profiles) and $d = 14$ (120 profiles), $m \leq 6$.

4 Complete proof for $d = 2$

The two ρ -orbits of $\mathcal{C}_2$ are $A = \{(1, 1, 0), (1, 0, 1), (0, 1, 1)\}$ and $B = \{(2, 0, 0), (0, 2, 0), (0, 0, 2)\}$. Reducing (1) by orbits (using Lemmas 3, 4 and dividing by $1 + q^m + q^{2m}$) gives

$$H_{A, m} = H_{A, m-1} + q^m H_{B, m-1}, \quad H_{B, m} = q^m H_{A, m-1} + (1 - q^m + q^{2m}) H_{B, m-1}. \quad (3)$$

Theorem 9 ($d = 2$). $H_{A,m} = \sum_{j \geq 0} q^{j^2} \begin{bmatrix} m \\ j \end{bmatrix}_q$ and $H_{B,m} = \sum_{j \geq 0} q^{j^2+j} \begin{bmatrix} m \\ j \end{bmatrix}_q$ (the classical Rogers–Ramanujan polynomials). Consequently $H_{\cdot,m} - H_{\cdot,m-1} \geq 0$ and $h_{c,m} \geq 0$ for all $c \in \mathcal{C}_2$, $m \geq 0$.

Proof. Both claimed forms hold at $m = 0$. Write $R_m^{(a)} = \sum_j q^{j^2+aj} \begin{bmatrix} m \\ j \end{bmatrix}_q$, $a \in \{0, 1\}$. The two q -Pascal rules $\begin{bmatrix} m \\ j \end{bmatrix}_q = \begin{bmatrix} m-1 \\ j \end{bmatrix}_q + q^{m-j} \begin{bmatrix} m-1 \\ j-1 \end{bmatrix}_q = q^j \begin{bmatrix} m-1 \\ j \end{bmatrix}_q + \begin{bmatrix} m-1 \\ j-1 \end{bmatrix}_q$ give

$$R_m^{(0)} - R_{m-1}^{(0)} = \sum_j q^{j^2+m-j} \begin{bmatrix} m-1 \\ j-1 \end{bmatrix}_q = q^m \sum_i q^{i^2+i} \begin{bmatrix} m-1 \\ i \end{bmatrix}_q = q^m R_{m-1}^{(1)},$$

which is the first equation of (3). For the second, the same q -Pascal rule gives (shifting $j = i + 1$, using $j^2 + j + (m - j) = j^2 + m$)

$$R_m^{(1)} - R_{m-1}^{(1)} = \sum_j q^{j^2+m} \begin{bmatrix} m-1 \\ j-1 \end{bmatrix}_q = q^m \sum_i q^{i^2+2i+1} \begin{bmatrix} m-1 \\ i \end{bmatrix}_q,$$

and the auxiliary identity $\sum_i q^{i^2+2i+1} \begin{bmatrix} m-1 \\ i \end{bmatrix}_q = R_{m-1}^{(0)} - (1 - q^m) R_{m-1}^{(1)}$ (proved by expanding both sides with q -Pascal and re-indexing; checked exactly in $\mathbb{Z}[q]$ for $m \leq 8$) yields $R_m^{(1)} = q^m R_{m-1}^{(0)} + (1 - q^m + q^{2m}) R_{m-1}^{(1)}$, the second equation of (3). Since $(H_{A,m}, H_{B,m})$ and $(R_m^{(0)}, R_m^{(1)})$ satisfy the same recursion (3) with the same initial values, they coincide.

Monotonicity is then manifest: $R_m^{(a)} - R_{m-1}^{(a)} = \sum_j q^{j^2+aj+m-j} \begin{bmatrix} m-1 \\ j-1 \end{bmatrix}_q \geq 0$. Proposition 7 yields $h_{c,m} \geq 0$ for $d = 2$. $\square$

5 The Q -transform

Theorem 10. With $Q_{n,c}$ defined by the Layer-1 D -tower (so $Q_{n,c} = D_n^n$),

$$Q_{n,c} = \sum_{m=0}^n (-1)^{n-m} q^{\binom{n-m}{2}} \begin{bmatrix} n \\ m \end{bmatrix}_q H_{c,m}.$$

Proof sketch. The tower $D_0^m = h_{c,m}$, $D_k^m = D_{k-1}^m - q^k D_{k-1}^{m-1}$ telescopes: by induction $D_k^m = \sum_j (-1)^j q^{\binom{j+1}{2}} \begin{bmatrix} k \\ j \end{bmatrix}_q h_{c,m-j}$ (standard q -binomial inversion), and substituting $h_{c,m} = H_{c,m} - (1 - q^m) H_{c,m-1}$ collapses the double sum by the q -Vandermonde/Pascal relations to the stated single alternating sum. Verified exactly against the D -tower for $d = 2, 4$ ($m \leq 6$, all profiles); for $d = 2$ it reproduces $Q_n = q^{n^2}$. $\square$

6 Bounded fermionic form conjecture

Conjecture 11 (Bounded A_2 Andrews–Gordon polynomials). For $\gcd(d, 3) = 1$ and each ρ -orbit O of $\mathcal{C}_d$ there is a fermionic form

$$H_{O,m} = \sum_{n_1, \dots, n_r \geq 0} q^{n^T B n + a^T n} \begin{bmatrix} m \\ n_1 \end{bmatrix}_q \prod_{i \geq 2} \begin{bmatrix} L_i(n) + \varepsilon_i \\ n_i \end{bmatrix}_q,$$

with B the A_2 /ASW quadratic form and all m -dependence confined to the single factor $\begin{bmatrix} m \\ n_1 \end{bmatrix}_q$.

Any such form gives Conjecture 8 immediately by q -Pascal ($\begin{bmatrix} m \\ n_1 \end{bmatrix}_q - \begin{bmatrix} m-1 \\ n_1 \end{bmatrix}_q = q^{m-n_1} \begin{bmatrix} m-1 \\ n_1-1 \end{bmatrix}_q \geq 0$), hence $h_m \geq 0$. *Evidence:* exact matches for both orbits at $d = 2$ ($m \leq 8$); for 3 of 5 orbits at $d = 4$ with $H = \sum q^{n_1^2+n_2^2-n_1n_2+an_1+bn_2} \begin{bmatrix} m \\ n_1 \end{bmatrix}_q \begin{bmatrix} 2n_1+\varepsilon \\ n_2 \end{bmatrix}_q$ ($m \leq 6$); for 4 of 7 orbits at $d = 5$ (CDU quadruple sums, $m \leq 5$). Evaluations $H_{c,m}(1) = ((\binom{d+2}{2})/3)^m$ ($= 2^m, 5^m, 7^m$ for $d = 2, 4, 5$) are consistent with the ASW mod-7 and CDU mod-8 families. Warnaar explicitly lists finding bounded analogues of the A_2 Andrews–Gordon identities as an open problem; $H_{c,m} = (q; q)_m F_{c,m}$ is the natural candidate for their left-hand sides. The unmatched orbits $((0, 2, 2), (0, 3, 1)$ at $d = 4$; $(0, 2, 3), (0, 3, 2), (0, 4, 1)$ at $d = 5$) resisted prefactor and two-term ansätze and likely require shifted binomials as in Warnaar’s Theorem 3.